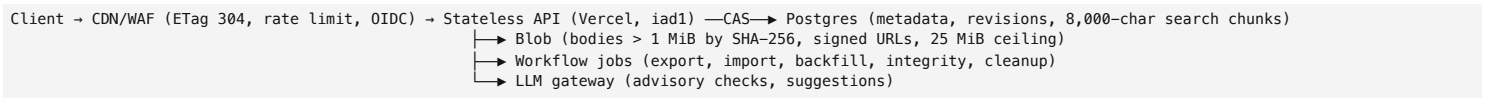
CLAUSE: production design

Where it stands: the deployed demo is open, shared, and fictional. Hosted runs on Vercel functions (`iad1`), Supabase Postgres (`us-east-1`) and private Vercel Blob; local runs on SQLite and the filesystem. Day-to-day operation (jobs, migrations, recovery, telemetry) is in `docs/OPERATIONS.md`.

Working assumptions (mine, not a brief): contracts around 1 MiB with a 25 MiB tail, hundreds of workspaces, up to 50 concurrent editors each, reads far above writes, 99.9% availability goal (unproven), US and EU customers.

Every number is tagged. **Measured** says where: a laptop (Apple M4, Node 24, local Postgres 16 or SQLite; numbers swing ~2× with background load) or **hosted** (Vercel `iad1` plus the database of the day, driven from a laptop over the public internet). **Target** is what production must hit. **Untested** means nobody has run it.



What the prototype measured

Operation	Measured	Target	What it tells us
Edit engine, 1 replacement in 10 MiB (laptop)	**7 ms**	n/a	The algorithm is not the cost; a 100-change batch is 9 ms
Word search, 400 KB, 8 clients, p95 (laptop Postgres, Oct 19)	**103 ms** (was 1,626)	< 300 ms	The chunk query rewrite removed a per-candidate lookup; machine heavily loaded, diagnostic
Edit, 400 KB, 8 clients, p95 (laptop Postgres, Oct 19)	**246 ms** (was 409)	< 500 ms	100 of 100 commits; same diagnostic caveat
Metadata read, 1 MiB, 8 clients, p95 (hosted, Neon, Oct 18)	**121 ms**	< 100 ms (**missed**)	Includes public-network latency; generator load 12→125
Word search, 1 MiB, 8 clients, p95 (hosted, Neon, Oct 18)	**348 ms**	< 300 ms (**missed**)	Measured before the query rewrite above
Edit, own 1 MiB document, 8 clients, p95 (hosted, Neon, Oct 18)	**268 ms**, ~27 commits/s	< 500 ms (met)	Each client edits its own document; no errors. Same-document contention is measured separately as CAS conflicts
32 clients (hosted, Neon)	errors in search and mixed load	no errors	No capacity claim at 32
25 MiB edit / revert / full replacement (hosted, single run)	**2.4 / 3.0 / 2.8 s**	publish separately	Functional check over Blob; API JSON stayed under 1 KB
LLM judge call	**~2,400 ms** median	hard deadline 25 s	Always advisory; the benchmark run cost $1.52

Sizing arithmetic (from the assumptions, not a load test): 500 concurrent editors at one edit per 15 s is ~33 edits/s. One Postgres committed ~27 separate-document edits/s at 8 clients, so production needs headroom: per-document row locks and more than one instance. **No hosted number above was taken on Supabase**; the matrix must be rerun there.

Decisions by topic

Topic	Design	Number	Trade-off
Compute	Stateless API on Vercel Fluid functions; slow work as durable Workflow jobs, at most 2 running, 100 queued	jobs: 3 retries, execution leases, 24 h export expiry	Two runtimes to observe vs. no request holding a 25 MiB export
Storage	Postgres: metadata, revisions, search chunks. Bodies above the 1 MiB inline limit live once in private Blob, keyed by SHA-256	25 MiB ceiling; duplicate DB bodies pruned only after a verified manifest and restore rehearsal	Two stores to keep consistent: integrity sweep and orphan cleanup run daily
Search	GIN over 8,000-character chunks (lifts the ~1 MB `tsvector` cap), `ts_rank`; `pg_trgm` for exact, case-sensitive matches; snippets only for the returned page	word p95 1,626 → 103 ms (laptop)	Chunk-level ranking, not whole-document
Index freshness	Chunks reindexed in the same transaction as the edit; large indexing has its own statement deadline	a committed edit is searchable at once	Long transactions on 25 MiB edits (~2.4 s)
Caching	Versions are immutable: cache by `(id, version/ETag)` at the edge; content by hash; Redis later only for rate limits and hot search pages	repeat reads `304`	Rate limits are per process today
Concurrency	Version compare-and-swap, `If-Match`, signed upload intents bound to `expectedVersion`; no auto-merge	stale write → `409`/`412`	Users retry; CRDTs would change the model
Batching	A PATCH batch is the atomic unit (5,000 changes); imports and backfills batched in jobs	100 edits = 9 ms engine	A large batch holds the document longer
CI/CD	Typecheck, lint, coverage floors, SQLite and Postgres suites, browser + a11y, weekly fuzz, audit; migrations are explicit commands, never on startup; `/ready` gates traffic	engine ≥98% lines, API ≥90%	Slower pipeline vs. fewer bad deploys
Security	OIDC/SSO + roles; `tenant_id` + row-level security; app connects as a scoped DB role over CA-verified TLS; audit of actor, version, hashes (never text)	demo has none (open, labelled)	Per-tenant keys add key management; SOC 2/GDPR need audited evidence
Resilience	Managed Postgres backups + PITR; Blob content covered by a recovery manifest and a restore rehearsal	RPO ≈ 0 in region, RTO < 5 min (targets)	One writer per document, so no active-active
Observability	OTLP metrics: route, SQL, pool, lock waits, Blob, AI cost, queue age, event loop; no text or IDs in labels	alerts: 5xx > 2%, queue age > 5 min, readiness down 2 min	Local collector verified; hosted delivery untested
Cost & regions	Cap document and batch size; per-IP AI limits (20 suggestions / 60 checks per hour); right-size compute from p95; regional cells for residency	current compute allows 60 Postgres connections	Cells duplicate infra; multi-region writes deferred

Risks and first measurements

1. **Hosted targets are not demonstrated.** At 8 clients, metadata (121 ms) and word search (348 ms) missed on Neon, before the search rewrite; 32 clients produced errors. Rerun the matrix on Supabase from a generator near `iad1`.
2. **Connection budget.** 60 Postgres connections on the current compute. The test suite exhausted the pooler when pointed at production; tests and load need their own database, and production headroom is untested.
3. **No tenancy today.** Nothing here is customer-ready until authentication, row-level security and audit are in; SOC 2 and GDPR are evidence and review, not architecture.

Raw numbers: October 18 and October 19 reports, `bench/performance`. Longer alternatives: appendix.